

Packaging FOSS Python applications and libraries for Debian

Codethink

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Debian

- Free and open-source software project producing a general-purpose Linux OS
- ~ 1400 contributors bound by the Debian Social Contract to serve **millions** of users and **thousands** of FOSS upstreams
- Debian Linux first released in 1993, and new releases (approximately) every 2 years thereafter
- A release is a **curated**, **well-behaved** and **tested** repository of **FOSS** software packages in the `.deb` format
- All the packages in a release are supported for at least **5 years**
- Debian packages are **redistributed** by derivatives (Ubuntu, Linux Mint, Kali Linux, Raspberry Pi OS, etc.)
- Debian packages are maintained by individuals and teams, who are likely **not upstream maintainers**

Installing a Python package on Debian

- **PyPi.org** hosts **binary** Python packages provided by upstream maintainers
- `pip` blocks global installs from PyPi in “managed environments”

```
error: externally-managed-environment

× This environment is externally managed
╰─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Debian-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have python3-full installed.

    If you wish to install a non-Debian packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.

    See /usr/share/doc/python3.13/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation
or OS distribution provider. You can override this, at the risk of breaking your
 Python installation or OS, by passing --break-system-packages.
```

- Instead, Debian packaging of upstream should be installed aiming for better system **integration**
- The Debian package may be out-of-date compared to upstream, but...
- The Debian package is more likely to:
  - Integrate well with **system utilities** and **libraries**
  - Integrate well with **dependencies** and **as a dependency**
  - Be supported with **bug fixes** and **security updates**
  - Have been **reviewed** by more people and processes
- In **high-assurance environments**, these qualities may matter more than latest updates

Apache BuildStream

- A Python program for **build integration**
- Used by Flatpak (Freedesktop-SDK), GNOME, and KDE communities to reproducibly build operating system images
- Supports caching of built artifacts using the **Remote Execution API (REAPI)**
- Upstream release was v2.7.0, Debian was on v1.6.9
- Upstream's 2.x series is more complex, with several non-Python dependencies

https://buildstream.build

Building the Debian package

- Debian is a **binary distribution**

- Installing a `.deb` package should put everything in the right place for the user:
  - Binaries
  - Configuration files
  - Man pages
  - Documentation

- Yet, Debian only accepts package uploads from maintainers in **source form**

- The `devscripts` package provides maintainers with tools to **build**, **test**, and **validate** their packages

- Packaging is configured with files under **`debian/`** directory added to upstream source tree

- Packaging entails **preparing `debian/` directory** so that the central build and quality-assurance systems can prepare a sound binary package for users

- 90% of Debian packaging is prepared in version control on **salsa.debian.org** (GitLab)

- Use of version control is not mandatory – maintainers have responsibility

The `debian/` directory

- Contains files which control local and remote Debian packaging software
- Mostly in the **RFC 822** format
- Can be generated by `debmake` from upstream sources for a new package – but needs careful manual checking
- Can be somewhat “magic” – but magic maintains consistency between packages
- For Python software with a `pyproject.toml`, and all dependencies in Debian, can be quite minimal

- **`debian/changelog`** – details Debian releases
- **`debian/control`** – package metadata
- **`debian/copyright`** – FOSS licence and attribution for **every file in the source distribution**
- **`debian/rules`** – Makefile for the build system
- **`debian/upstream/metadata`** – links to upstream VCS, bug trackers, and mailing lists
- **`debian/patches`**

debian/changelog

```
buildstream (2.8.1-1) unstable; urgency=high

  * New upstream release
    - SECURITY UPDATE: Improper link resolution before file access
      ('link following') vulnerability in the `tar` source plugin
      when running on Python < 3.12
    - CVE-2026-82331
  * d/copyright: update for upstream 2.8.1

 -- Theodore Tucker <theodore@tti0.net>  Wed, 23 Sep 2026 08:25:08 +0100
```

debian/control (abridged)

```
Source: buildstream
Standards-Version: 4.7.4
Maintainer: Debian GNOME Maintainers <pkg-gnome-maintainers@lists.alioth.debian.org>
Section: gnome
Build-Depends: debhelper-compat (= 14),
               dh-sequence-python3,
               cython3,
               pybuild-plugin-pyproject,
               python3-all,
               python3-all-dev,
               python3-click,
                <snip>
Build-Conflicts: native-architecture-is-s390x,
Vcs-Browser: https://salsa.debian.org/gnome-team/buildstream
Vcs-Git: https://salsa.debian.org/gnome-team/buildstream.git
Homepage: https://buildstream.apache.org

Package: python3-buildstream
Architecture: any
Multi-Arch: foreign
Section: python
Depends: buildbox,
Recommends: lzip,
Description: Python library for BuildStream toolset
 <snip>
```

debian/copyright (abridged)

```
<snip>

Files: *
Copyright: 2017-2026 The Apache Software Foundation
License: Apache-2.0

Files: doc/bst2html.py
Copyright: 2013 German M. Bravo (Kronuz)
           2018 Codethink Limited
License: Expat

<snip>
```

debian/rules (abridged)

```
#!/usr/bin/make -f
export DEB_BUILD_MAINT_OPTIONS = hardening=+all
export PYBUILD_TEST_PYTEST = 0

%:
	dh $@ --buildsystem=pybuild

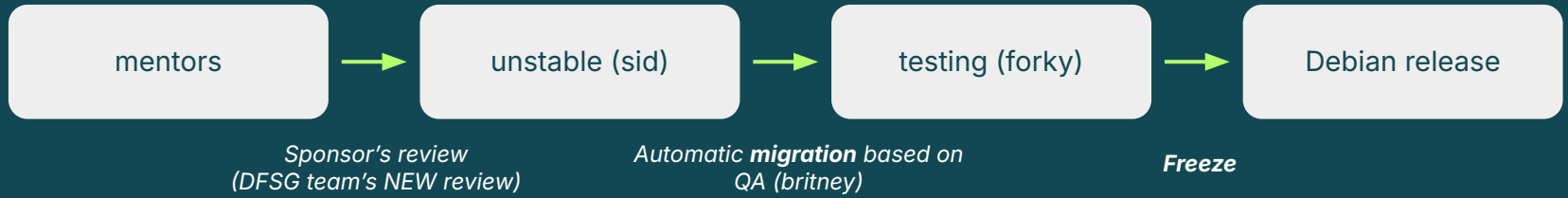
# BuildStream ships with pre-compiled Python GRPC protocols
# These are incompatible with Debian's outdated python3-protobuf version, so we
# need to recompile with that version
execute_before_dh_auto_build:
	python3 setup.py build_grpc

<snip>
```

From upload to release

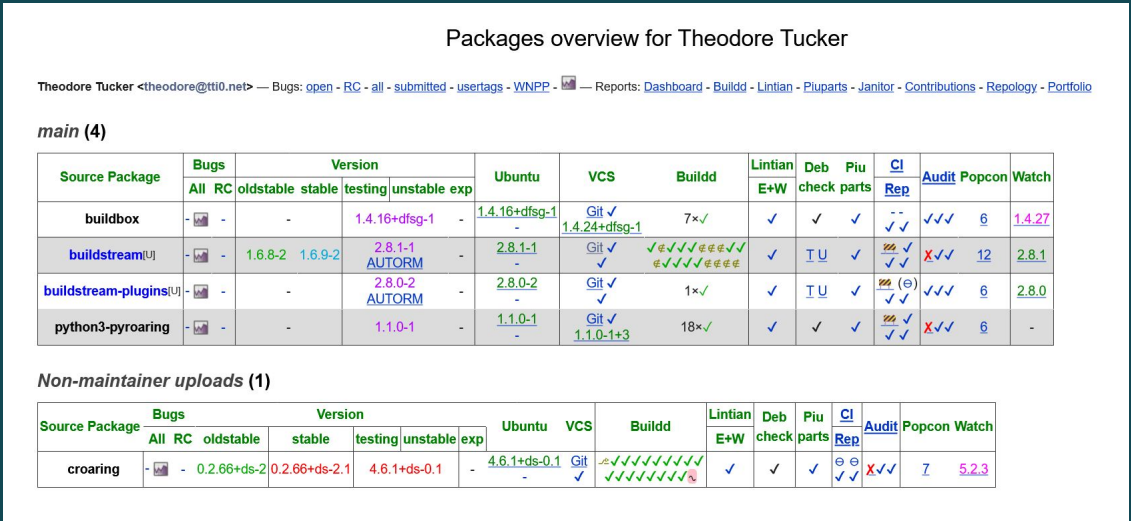
- **sbuild** used to build the package according to `debian/rules` in a local **chroot**, producing a `.changes` file
- `.changes` file is **PGP-signed** *(debsign)* and **uploaded by FTP** *(dput)*

- **Sponsored maintainers** can upload to **mentors.debian.net**
- **Debian Maintainers** can upload specific packages to **ftp-master.debian.org**
- **Debian Developers** can upload any package to **ftp-master.debian.org**



Quality assurance

- Debian's **automated quality assurance processes** are **very extensive**
- QA takes place **after upload**, but **before migration** to `testing`
- **Failed QA blocks migration** or causes **auto-removal** if already in `testing`
- QA checks can be **emulated** in GitLab CI on salsa.debian.org



Packages overview for Theodore Tucker

Theodore Tucker <theodore@tti0.net> — Bugs: open - RC - all - submitted - usertags - WNPP - — Reports: Dashboard - Buildd - Lintian - Piuparts - Janitor - Contributions - Repology - Portfolio

***main* (4)**

Source Package	Bugs		Version					Ubuntu	VCS	Buildd	Lintian	Deb	Piu	CI	Audit	Popcon	Watch
	All	RC	oldstable	stable	testing	unstable	exp				E+W	check	parts	Rep			
buildbox	-	-	-		1.4.16+dfsg-1		-	1.4.16+dfsg-1 -	Git ✓ 1.4.24+dfsg-1	7×✓	✓	✓	✓	- - ✓ ✓	✓✓✓	6	1.4.27
buildstream[U]	-	-	1.6.8-2	1.6.9-2	2.8.1-1 AUTORM		-	2.8.1-1 -	Git ✓ ✓	✓✓✓✓✓✓✓✓✓✓✓✓✓✓	✓	T U	✓	✓ ✓	X✓✓	12	2.8.1
buildstream-plugins[U]	-	-	-		2.8.0-2 AUTORM		-	2.8.0-2 -	Git ✓ ✓	1×✓	✓	T U	✓	(⊖) ✓ ✓	✓✓✓	6	2.8.0
python3-pyroaring	-	-	-		1.1.0-1		-	1.1.0-1 -	Git ✓ 1.1.0-1+3	18×✓	✓	✓	✓	✓ ✓	X✓✓	6	-

***Non-maintainer uploads* (1)**

Source Package	Bugs		Version					Ubuntu	VCS	Buildd	Lintian	Deb	Piu	CI	Audit	Popcon	Watch
	All	RC	oldstable	stable	testing	unstable	exp				E+W	check	parts	Rep			
croaring	-	-	0.2.66+ds-2	0.2.66+ds-2.1	4.6.1+ds-0.1		-	4.6.1+ds-0.1 -	Git ✓	✓✓✓✓✓✓✓✓✓ ✓✓✓✓✓✓✓✓	✓	✓	✓	⊖ ⊖ ✓ ✓	X✓✓	7	5.2.3

Quality assurance

- **buildd** – produces architecture-specific binary packages and uploads to archives
- **lintian** – checks `debian/` directory for policy compliance
- **uscan** – checks for new upstream releases
- **debaudit** – checks Debian source code matches upstream repository
- **piuparts** – checks end-users can install and uninstall the package cleanly
- **autopkgtests** – integration tests of the package in-situ
- **reproduce.debian.net** – checks binary packages in archive can be reproduced bit-for-bit from source upload
- **reproducible-builds.org** – reproducibility stress test
- **bugs.debian.org** – reports from users and other Debian contributors

Responsibilities of a maintainer

- Respond to bug reports, forwarding patches upstream if appropriate
- Responding to auto-removals
- Check for new upstream releases and update packaging according to policy
- Follow upstream news
- Take care of your reverse dependencies – no ABI breaks are allowed without an official **transition**
- Fix bugs in your dependencies' packaging

Sources of help in Debian

- **Debian Policy Manual** – official rules for packages
  https://www.debian.org/doc/debian-policy/
- **Debian Python Policy** – official rules for Python packages
  https://www.debian.org/doc/packaging-manuals/python-policy/
- **Guide for Debian Maintainers** – practical instructions for preparing a `debian/` directory
  https://www.debian.org/doc/manuals/debmake-doc
- **Python Library Style Guide** – practical instructions for packaging Python libraries
  https://wiki.debian.org/Python/LibraryStyleGuide
- **Debian Mentors FAQ** – advice on the social aspects of your first contributions
  https://wiki.debian.org/DebianMentorsFaq

Conclusion – responsibility and reward

– **Responsibility** and **quality** are the watchwords

– Packaging your Python software for Debian helps you:

  – Reach more users in high-assurance environments
  – Understand how your software fits into a system
  – Learn from experts to follow best practices upstream
  – Strive to give your users a stable and polished experience

– UK Government's DRIVE35 project has contributed to the Debian packaging of BuildStream 2.x and dependencies

https://gitlab.com/theodoretucker_ct/slides



Thank You.
Questions?

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